Exact Synthesis of Two-Qubit Clifford+CS Circuits with a Near-Minimal Number of K and CS Gates

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We present a method for decomposing an arbitrary two-qubit Clifford+CS operator (equivalently, a 4×4 unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$) into a Clifford+CS circuit without ancillae. Given a unitary U, let k be the least natural number such that all entries of $(1+i)^k U$ are Gaussian integers. Our method decomposes U, up to a global phase, into a circuit of length at most $20k + 9$, using at most $2k$ K gates (i.e., scaled H gates) and at most $2k+1$ Controlled-S gates. The K-count is at most 2 times the optimal, and the CS-count is at most 4 times the optimal. In experiments, our method always achieves the optimal CS-count and yields a mean K-count reduction of 19% compared to the best known result. Our approach is similar to that of Russell [35] for the Clifford+T gate set, but provides explicit bounds (e.g., $20k + 9$ vs. $O(k)$) and operates over the simpler ring $\mathbb{Z}[\frac{1}{2}, i]$, a subring of the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ used by Russell.

I. INTRODUCTION

In quantum computation, decomposing unitary operators into gates from a fixed universal set is a fundamental problem [33]. The Clifford+Controlled-S gate set (abbreviated Clifford+CS) is universal, just like the well-known Clifford+T gate set. We study the exact synthesis of two-qubit Clifford+CS operators. Previous work on two-qubit synthesis of Clifford+CS or Clifford+T operators has mainly focused on minimizing the CS- or T-count, since these gates are more expensive than Clifford operators in certain scenarios. However, work on T-gate optimization [1, 5, 9, 10, 21, 25] has suggested that the H gate is an "obstruction" to T-optimization methods. Quoting from [1]: "It has been shown that T-depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly".

We propose a method that produces circuits with a near-minimal number of both K (i.e., scaled H) gates and CS gates, thereby alleviating this obstruction. The problem of simultaneously minimizing two different gate counts is also of mathematical interest: previous work on single-qubit Clifford+T synthesis achieved the minimum use of both the Hadamard and T gates [14, 27, 31], and extending this to the multi-qubit case has been an open problem.

A. Background

Consider the following unitary operators:

$$i,\ K = \omega^7 H = \frac{1}{1+i}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix},\ S = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix},\ \text{and}$$

$$CZ = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1\end{bmatrix}.$$

Here, i is the complex unit; $\omega = e^{\frac{i\pi}{4}}$ is a primitive 8th root of unity; H is the Hadamard gate and K is a scaled version of H; S is sometimes called the *phase gate*; and CZ is the controlled-Z gate. Under multiplication, identities, and tensor products, these operators generate the *Clifford group*. (The precise group depends on which scalars are included; we include only the scalars ± 1 and $\pm i$, so the above gates generate the Clifford group exactly.)

Every operator U obtained in this way has size $2^n \times 2^n$ for some natural number n, and we say that U acts on n qubits. The group of n-qubit Clifford operators is finite for any given n (see, e.g., [36]), and therefore not universal for quantum computing. Universality is obtained by adding the controlled phase gate

$$CS = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & i\end{bmatrix}.$$

The resulting operators are called Clifford+CS operators. Note that $CZ = CS^2$. We write $\mathcal{CCS}$ for the group of two-qubit Clifford+CS operators and $\mathcal{CIR}$ for the set of two-qubit Clifford+CS circuits (a circuit being a finite sequence of gates). Each circuit $D \in \mathcal{CIR}$ implements a unique operator in $\mathcal{CCS}$, denoted $[D]$. That is, $[_]$ is the matrix semantics function from $\mathcal{CIR}$ to $\mathcal{CCS}$.

We use the following notation: $K_0 = K \otimes I$, $K_1 = I \otimes K$, and similarly for S_0 and S_1, where I is the 2×2 identity operator. We identify the scalar i with the 4×4 matrix $i\,(I \otimes I)$. For controlled gates, the last index denotes the target. For example, CX_{01} is the controlled-X gate with control qubit 0 and target qubit 1, shortened to CX; CX_{10} is shortened to XC. The controlled-S and controlled-Z gates are symmetric: $CS_{jk} = CS_{kj}$

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and $CZ_{jk} = CZ_{kj}$, so we simply write CS and CZ. We also use the Z-gate, the X-gate, the controlled-K gate $CK = I \oplus K$, and the swap gate Ex. These are related to the primitive gates by $Z = S^2$, $X = KZKi$, $CX = K_1 CZ K_1 \cdot i$, $Ex = CX\,XC\,CX$, and

$$CK = CZ \cdot Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

As with CX, we shorten CK_{01} to CK and CK_{10} to KC. By Equation 1, $CK \in \mathcal{CCS}$.

B. Related work

Operator synthesis can be performed exactly or approximately. We restrict attention to exact synthesis and list the most closely related works.

1. Single-qubit and single-qudit

For single-qubit Clifford+T synthesis, the works [14, 27, 31] achieved the minimum use of both the Hadamard and T gates. The works [16, 34] achieved the minimum T-count for single-qutrit Clifford+T synthesis. The work [23] considered single-qudit Clifford+T synthesis. Forest et al. [12] generalized exact synthesis from Clifford+T to single-qubit Clifford-cyclotomic gate sets. Kliuchnikov and Yard [29] developed a general algebraic framework for exact synthesis over number rings, unifying several gate sets under one theory.

2. Two-qubit

Two works on two-qubit synthesis guarantee optimality of a non-Clifford gate: Li et al. [30] for Clifford+T (near T-optimal, with worst-case T-count at most 10 times the minimum) and Glaudell et al. [17] for Clifford+CS (CS-optimal). Neither considered H-count optimization, but for the latter, we can deduce that its K-count is at most 2 times the optimal (the same bound as ours). Glaudell et al. [15] later gave a high-performance exact synthesis framework for two-qubit Clifford+T circuits that optimizes the T-count exactly, using a meet-in-the-middle strategy with algebraic canonicalization and lookup tables. All three of these works rely on the exceptional isomorphism $SU(4) \cong \mathrm{Spin}(6)$. By contrast, our work does not use this isomorphism. Instead, we employ a residue analysis similar to that of Russell [35], but over the simpler ring $\mathbb{Z}[\frac{1}{2}, i] \subset \mathbb{Z}[\frac{1}{\sqrt{2}}, i]$. Given a unitary U, Russell's method outputs a circuit of gate count $O(k)$, where k is the least natural number such that the entries of $(1+\omega)^k U$ lie in $\mathbb{Z}[\omega]$.

3. Multi-qubit and multi-qutrit

Multi-qubit Clifford+T synthesis was introduced by Giles and Selinger [13], whose algorithm has worst-case gate count $O\left(3^{2^n} nk\right)$. This was later improved [26] to $O\left(4^n nk\right)$. Glaudell et al. gave exact multi-qubit synthesis for several subclasses of Clifford+T operators [4], such as the Toffoli+Hadamard operators. Amy et al. adapted the method of [26] to Toffoli+Hadamard operators, achieving a similar improvement in gate count [3]. None of these multi-qubit synthesis works consider T-count optimality. Kliuchnikov and Schönnenbeck [28] recently developed an optimal search algorithm for multi-qubit exact synthesis using A^* search; for two-qubit circuits, its runtime is proportional to the T-count. The works [2, 11] considered exact synthesis of multi-qubit Clifford+Cyclotomic operators, while [19, 24] treated the multi-qutrit case. The work [18] proved that qutrit metaplectic gates are a subset of Clifford+T. The work [32] proposed a heuristic algorithm achieving V-gate optimality.

II. SOME ALGEBRA

A. Rings

We write $\mathbb{N} = \{0, 1, \ldots\}$ for the natural numbers and $\mathbb{Z}$, $\mathbb{R}$ for the integers and real numbers, respectively. We write $z^\dagger$ for the complex conjugate of a complex number z. Recall that a *unit* in a ring is an invertible element.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. It is a Euclidean domain [6], so division with remainder, greatest common divisors, divisibility, and the notion of a prime (i.e., a non-unit that cannot be written as a product of two non-units) are all available in $\mathbb{Z}[i]$. Let $\gamma = 1+i$. It is well-known that γ is a Gaussian prime [22]. Since $\gamma^2 = 2i$, the elements γ^2 and 2 divide each other. Using Euclidean division, one can verify that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1+i, 1-i, 2\}.$$

The above is a slight abuse of notation: strictly, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal (γ), whose elements are residue classes $x \bmod (\gamma)$ for $x \in \mathbb{Z}[i]$. We omit $\bmod (\gamma)$ for brevity. The sets $\{0, 1\}$ etc. are complete sets of residue class representatives, coinciding with the remainders from Euclidean division by γ.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider the ring $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. Every $t \in \mathbb{D}[i]$ satisfies $\gamma^k t \in \mathbb{Z}[i]$ for some $k \in \mathbb{N}$, motivating the following definition.

Definition II.1. Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for t if $\gamma^k t \in \mathbb{Z}[i]$. The least such k is called the *least denominator exponent* of

t, denoted by $\mathrm{lde}(t)$. Equivalently, the least denominator exponent of t is the smallest $k \in \mathbb{N}$ such that t can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that k is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least k that is a denominator exponent for all of its entries.

Remark II.2. By enumerating all 2-qubit Clifford operators, we have that the lde of a 2-qubit Clifford operator is at most 2.

B. Residue rings

The following characterization of the quotient rings is well-known:

$$\mathbb{Z}[i]/(\gamma^n) = \left\{ \sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0,1\} \right\}.$$

We give an elementary proof by induction on n; a similar argument for the case $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ appears in Clement [8]. The base case $n = 1$ is immediate. Suppose the claim holds for n, i.e., every $x \in \mathbb{Z}[i]$ is congruent to some $\sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n)$, meaning

$$x = \sum_{i=0}^{n-1} b_i \gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0,1\}$, substituting and simplifying gives

$$x = \sum_{i=0}^{n} b_i \gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i \gamma^i \bmod (\gamma^{n+1}).$$

For brevity, we omit the $\bmod (\gamma^n)$ in $\sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n)$. Since $b_i \in \{0,1\}$, we write the binary string $(b_0 b_1 ...)_2$ for the sum; the parentheses and subscript $_2$ may be omitted when the context makes clear that $b_0 b_1 ...$ denotes a binary residue.

Moreover, residue classes are distinct: if $b_0 b_1 \dots b_{n-1}$ and $b'_0 b'_1 \dots b'_{n-1}$ are congruent, i.e., $\sum_{i=1}^{n-1} b_i \gamma^i - \sum_{i=1}^{n-1} b'_i \gamma^i = k\gamma^n$ for some k, then $b_i = b'_i$ for all i. Indeed, reducing both sides modulo (γ) gives $b_0 = b'_0$. Cancelling b_0 and b'_0 and dividing by γ yields

$$\sum_{i=1}^{n-1} b_i \gamma^{i-1} - \sum_{i=1}^{n-1} b'_i \gamma^{i-1} = k\gamma^{n-1}$$

Reducing modulo (γ) again gives $b_1 = b'_1$. Repeating this argument shows $b_i = b'_i$ for all i.

We write $\rho_n(x)$ for the canonical projection from $\mathbb{Z}[i]$ to n-digit residues. The proof above shows that $\rho_n(x)$ is *stable* under increase of n: the first n digits of $\rho_{n+1}(x)$ coincide with $\rho_n(x)$. Moreover, ρ_2 is stable under complex conjugation: $\rho_2(x^\dagger) = \rho_2(x)$. We sometimes write $x \equiv_{\gamma^n} y$ as shorthand for $x \equiv y \bmod \gamma^n$. Modulo γ^3, we have

x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$
1	100	-1	101	i	111	$-i$	110
0	000	$1+i$	010	$1-i$	011	2	001

The $(\bmod\ \gamma^n)$ residue classes form a commutative ring. Modulo γ^2, the ring addition coincides with bitwise exclusive or (denoted $\oplus$); $(10)_2$ is the multiplicative identity; $(00)_2$ is the zero element; and multiplication by $(01)_2$ is the right shift operation. By distributivity, $11 \cdot b_0 b_1 = 10 \cdot b_0 b_1 + 01 \cdot b_0 b_1 = b_0(b_1 \oplus b_0)$. Since $\rho_2(i) = 11$, whenever $\rho_1(x) = 1$, either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo γ^3, the addition rule is

$$b_0 b_1 b_2 + b'_0 b'_1 b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where $\cdot$ denotes multiplication in $\mathbb{Z}[i]/(\gamma)$, equivalently the logical "and" of two bits. The multiplicative identity is 100 and the zero element is 000. Multiplication by 010 is a right shift by 1 digit and multiplication by 001 is a right shift by 2 digits. The rest of the multiplication table follows by distributivity. The additive inverse is

$$-(b_0 b_1 b_2) = b_0 b_1 (b_2 \oplus b_0).$$

Inspecting the ρ_3 table, we see that ρ_3 is not stable under complex conjugation; instead, $\rho_3(x^\dagger) = \rho_3(x) + 001$. Since $\rho_3(i) = 111$, whenever $\rho_1(x) = 1$, there exists $k \in \{0,1,2,3\}$ such that $\rho_3(i^k x) = 100$.

Definition II.3. For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say x is *even*, otherwise *odd*.

For example, $2+3i$ and $5+2i$ are odd, while $2+4i$ and $1+3i$ are even. In our convention, an even Gaussian integer is one divisible by γ, not necessarily by 2. However, for $x \in \mathbb{Z}$, x is even (resp. odd) as an integer if and only if it is even (resp. odd) as a Gaussian integer, so our definition extends the usual notion of parity on the integers. The following lemmas are straightforward.

Lemma II.4. *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then α is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

Lemma II.5. *If $\alpha \in \mathbb{Z}[i]$ is odd, then $\alpha \equiv \pm 1, \pm i \,(\bmod\ \gamma^3)$. In other words, $\alpha \equiv i^e \,(\bmod\ \gamma^3)$ for some $e \in \{0,1,2,3\}$. Moreover, $\alpha \equiv i^e \,(\bmod\ \gamma^2)$ for some $e \in \{0,1\}$.*

Lemma II.6. *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Let k be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$, we define the residue function $\rho_n^k(x) = \rho_n(\gamma^k x)$. We call $\rho_n^k(x)$ the (n,k) residue of x. If k is a denominator exponent for a vector or matrix A, $\rho_n^k(A)$ is defined entry-wise.

Lemma II.7. *For $t \in \mathbb{D}[i]$, if $k = lde_\gamma(t) > 0$, then $\gamma^k t$ is odd.*

Lemma II.8. *For any $x \in \mathbb{D}[i]$ with denominator exponent k, $\rho_2^k(x^\dagger) = \rho_2^k(x)$ and $\rho_3^k(x^\dagger) = \rho_2^k(x) + 001$.*

C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$; then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say i acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$. Given an n-digit residue $b_0 b_1 \dots b_{n-1}$, a *right shift* produces the $(n+1)$-digit residue $0 b_0 b_1 \dots b_{n-1}$. Given an $(n+1)$-digit residue $0 b_0 b_1 \dots b_{n-1}$, a *left shift* produces $b_0 b_1 \dots b_n$. We only allow shifts that are "well-defined" in the sense that no information is lost and no unknown digit is introduced. (The usual right shift was used above when describing multiplication in $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$.) We write $LS(x)$ for the left shift of x and RS for the right shift.

Lemma II.9. *For $t \in \mathbb{D}[i]$, k a denominator exponent of t, we have*

$$\rho_{n+1}^{k+1}(t) = \rho_{n+1}^k(\gamma t) = RS(\rho_n^k(t)) \text{ and}$$

and if $\gamma^k t$ is even,

$$\rho_n^{k-1}(t) = \rho_n^k(t/\gamma) = LS(\rho_{n+1}^k(t)).$$

Given $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we have $\rho_2^k(x) = 1b$ and $\rho_2^k(y) = 1b'$ for some $b, b' \in \{0,1\}$. Hence $\rho_2^k(x+y) = 1b + 1b' = 0(b \oplus b')$, and so $\rho_1^{k-1}(x+y) = LS(\rho_2^k(x+y)) = b \oplus b'$. Similarly, $\rho_1^{k-1}(x-y) = LS(\rho_2^k(x-y)) = LS(1b - 1b') = LS((1-1)(b-b')) = b \oplus b'$, since subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$ (every element is its own additive inverse). If $b \oplus b' = 1$, then lde$(x \pm y)$ decreases by 1; if $b \oplus b' = 0$ and $k \geq 2$, then lde$(x \pm y)$ decreases by at least 2. Since dividing t by γ increases the lde by 1, for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 10b$ and $\rho_3^k(y) = 10b'$, we obtain

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

Hence lde$(K \begin{bmatrix} x \\ y \end{bmatrix}) = k-1$, with one entry having lde $k-1$ and the other at most $k-2$ (provided $k > 2$).

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

P	circuit	P	circuit	P	circuit
0123	identity	1023	$X_1 \cdot CX$	2103	$X_0 \cdot XC$
1203	$Ex \cdot X_0 \cdot XC$	2013	$Ex \cdot X_1 \cdot CX$	0213	Ex
3210	$X_1 \cdot X_0$	2310	$X_0 \cdot CX$	2130	$Ex \cdot X_1 \cdot XC$
3120	$Ex \cdot X_1 \cdot X_0$	1320	$Ex \cdot X_0 \cdot CX$	1230	$X_1 \cdot XC$
3012	$XC \cdot X_1$	0312	$Ex \cdot XC$	0132	CX
3102	$Ex \cdot CX \cdot X_0$	1302	$Ex \cdot X_0$	1032	X_1
3021	$Ex \cdot XC \cdot X_1$	0321	XC	0231	$Ex \cdot CX$
3201	$CX \cdot X_0$	2301	X_0	2031	$Ex \cdot X_1$

TABLE I. Circuit implementation of 24 permutation matrices.

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1') \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \end{bmatrix}.$$

III. SPECIAL UNITARIES

A. Permutation matrices

Let $a_0a_1a_2a_3$ be a permutation of $\{0,1,2,3\}$. The permutation matrix $P_{a_0a_1a_2a_3}$ is the unitary matrix that maps e_i to e_{a_i}, where $\{e_i\}$ is the standard basis. All 24 permutation matrices are Clifford operators and can be implemented by circuits of length at most 3, see Table I.

B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ (and $\mathbb{Z}[i]$) are of the form

$$\text{diag}(i^a, i^b, i^c, i^d), \text{ where } a, b, c, d \in \{0,1,2,3\}.$$

They can all be implemented by Clifford+CS circuits with length of at most 6 up to a global phase i^k, using at most 1 CS gate. This is because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k, b_i \in \{0,1\}, k \in \{0,1,2,3\},$$

are all distinct and diagonal, and their number is 4^4, which equals the cardinality of the set of diagonal unitary matrices.

C. Generalized permutation unitary matrices

A *generalized permutation* (unitary matrix with entries in $\mathbb{D}[i]$) is a matrix having the same nonzero pattern as a permutation matrix, but whose nonzero entries may be arbitrary powers of i (i.e., unit-length elements of $\mathbb{Z}[i]$). Every 4×4 generalized permutation is a product of a permutation matrix and a diagonal unitary matrix, and hence can be implemented by a circuit of length at most 9 containing at most 1 CS gate and no K gate. The group of generalized permutations is a semidirect product of the permutation group and the diagonal unitary group. In particular, for any permutation matrix P and diagonal matrix D, there exists D' such that

$$PD = D'P$$

and the CS-count (see Definition IV.2) of D and D' are the same.

IV. THE EXACT SYNTHESIS ALGORITHM

Our method follows a residue analysis similar to Russell's [35] for the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$, but operates over the simpler subring $\mathbb{Z}[\frac{1}{2}, i]$. Such residue analysis methods are widely used in exact synthesis; see, e.g., [3, 13, 14, 17, 30]. Compared to the ad hoc use of residues in prior work, we have given in Section II a uniform treatment based on binary residues, which we conjecture generalizes to any ξ-ring as defined in [28].

We begin with the main theorem.

A. Main theorem

Theorem IV.1. *For any $A \in \mathcal{CCS}$ with lde k, there exists a circuit C implementing A, using at most $2k$ K gates, at most $2k+1$ CS gates, and at most $9(2k+1)+2k$ gates up to a global phase.*

The proof is constructive and yields an algorithm for computing such a circuit.

B. Optimality

Definition IV.2. Let D be a Clifford+CS circuit. Recall that $[D]$ is the unitary operator it implements. Define

- $\mathrm{rkc}(D) =$ the number of K gates that D contains.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D) =$ the number of CS gates that D contains.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D) =$ the number of gates that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- by abuse of notation, $\mathrm{cs}, \mathrm{kc}, \mathrm{len}$ are also defined on operators that are implementable by circuits.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ respectively. We call $\mathrm{rkc}(D)$ and $\mathrm{rcs}(D)$ the (raw) K-count and (raw) CS-count of D respectively, and $\mathrm{kc}(D)$, $\mathrm{cs}(D)$ the optimal K-count and the optimal CS-count of D respectively.

Remark IV.3. This convention differs from that of, e.g., [20], where $\mathrm{cs}(D)$ would be called the CS-count of D. In their convention, there is no separate notion of the "raw" K-count or CS-count; the "count" always refers to the optimal value.

Any Clifford+CS circuit D is an alternating sequence of Clifford parts C_i and a CS gate

$$D = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all two-qubit Clifford circuits, each C_i is implementable by at most 2 K gates and satisfies $\mathrm{lde}(C_i) \leq 2$. By the subadditivity of lde, $\mathrm{lde}(D) \leq \mathrm{lde}(C_0) + \mathrm{lde}(CS) + \mathrm{lde}(C_1) + \ldots + \mathrm{lde}(C_n) \leq 2(n+1)$, i.e., $\mathrm{lde}(D) \leq 2(\mathrm{rcs}(D)+1)$, which implies

$$\mathrm{lde}(D) \leq 2(\mathrm{cs}(D)+1).$$

If (2) is CS-optimal, i.e., $\mathrm{cs}(D) = n$, then $\mathrm{rkc}(D) \leq 2(\mathrm{cs}(D)+1)$, which implies

$$\mathrm{kc}(D) \leq 2(\mathrm{cs}(D)+1).$$

Any Clifford+CS circuit D can also be written as an alternating sequence of generalized permutations P_i and K_1 gates (by replacing every K_0 with $Ex \cdot K_1 \cdot Ex$):

$$D = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

Each generalized permutation uses at most 1 CS gate and no K gate, so $\mathrm{cs}(D) \leq \mathrm{rkc}(D)+1$, which implies

$$\mathrm{cs}(D) \leq \mathrm{kc}(D)+1.$$

By subadditivity of lde, $\mathrm{lde}(D) \leq \mathrm{rkc}(D)$, which implies

$$\mathrm{lde}(D) \leq \mathrm{kc}(D).$$

Theorem IV.1 produces D' with $\mathrm{rkc}(D') \leq 2\mathrm{lde}(D)$ and $\mathrm{rcs}(D') \leq 2\mathrm{lde}(D)+1$. Combined with the preceding inequalities, this gives

$$\mathrm{lde}(D) \leq \mathrm{kc}(D) \leq 2\mathrm{lde}(D) \tag{4}$$

$$\mathrm{kc}(D)/2 - 1 \leq \mathrm{cs}(D) \leq 2\mathrm{lde}(D)+1 \tag{5}$$

Thus our synthesis algorithm achieves at worst 2 times the optimal K-count and at worst 4 times the optimal CS-count. (Experiments show it always achieves the optimal CS-count and a strictly better K-count than the best known method; see Section V.) Inequality (5) is similar to Proposition 6 of [17], but slightly tighter.

1. K-optimality of Glaudell et al.'s algorithm

Although Glaudell et al. [17] did not consider the K-count, we can derive from their CS-optimality that for any unitary A,

$$\mathrm{cs}(A) \le \mathrm{kc}(A) \le 2\,\mathrm{cs}(A) + 2.$$

This follows because their output has the form of Equation (2) with optimal CS-count, and each Clifford part between two CS gates has K-count 1 or 2. Let E be such a Clifford part lying between two CS gates. First, $\mathrm{kc}(E) \neq 0$: if E were a generalized permutation (which contains no CS gate since E is Clifford), then the two flanking CS gates in $CS \cdot E \cdot CS$ could be cancelled, contradicting CS-optimality. The cancellation follows by direct calculation over the finitely many generalized permutations, or by the semidirect product structure of permutations and diagonals. That $\mathrm{kc}(E) \le 2$ follows from enumerating all two-qubit Cliffords.

Thus Glaudell et al.'s algorithm enjoys the same property as ours: the K-count is at most 2 times the optimal.

Remark IV.4. Note that experiments show that in most cases $\mathrm{lde}(A) < \mathrm{cs}(A)$ for randomly generated A. So our bound $\mathrm{kc}(A) \le 2\mathrm{lde}(A)$ is strictly better in practice.

C. Complexity

Each iteration reduces the lde by at least 1, using only finitely many operations (independent of the lde; proved below). Since unitaries with lde 0 are generalized permutations, the time complexity is $O(\mathrm{lde}(U))$ for any input $U \in \mathcal{CCS}$.

D. Lemmas and proofs

We begin with several lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of the complex numbers, the standard notions of complex conjugation (denoted $_^\dagger$), inner product, norm, transposition (denoted $_^T$), and conjugate transposition (also denoted $_^\dagger$) carry over. For a vector $v \in \mathbb{D}[i]^n$, the *norm* is $\|v\| = \sqrt{v^\dagger v}$, and $\|v\|^2 = v^\dagger v \in \mathbb{D}$ since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly, if $v \in \mathbb{Z}[i]^n$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1. For a vector v, the notation v_j refers to its jth entry; entries are numbered starting from $j = 0$.

Lemma IV.5. *For any $A \in \mathcal{CCS}$ with lde k, there exist permutation matrices L and R such that $\rho_1^k(LAR)$ or $\rho_1^k(LAR)^T$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

We say A has such a pattern *and write* $\mathrm{pat}(A)$ *for the pattern of A. Moreover, A has pattern (i) if and only if $\mathrm{lde}(A) = 0$, i.e., when A is a generalized permutation. Patterns (iv), (v), and (vi) occur only when $\mathrm{lde}(A) > 1$.*

Before proving this lemma, we prove several useful lemmas.

Lemma IV.6. *Let v be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$. There exists a generalized permutation matrix P such that $\rho_1^k(Pv) = (1,1,0,0)^T$ or $(1,1,1,1)^T$, $\rho_2^k(Pv) = (10,10,01,00)^T$ or $(10,10,10,10)^T$ when $k > 1$, and $\rho_2^1(Pv) = (10,10,00,00)^T$ when $k = 1$.*

Proof. Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show that w has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\|^2 = |\gamma|^{2k}\|v\|^2 = (\gamma^\dagger\gamma)^k \equiv 0 \,(\mathrm{mod}\ \gamma).$$

Therefore, an even number of indices $j \in \{0,\dots,3\}$ satisfy $\|w_j\|^2$ odd. By Lemma II.4, an even number of the w_j are odd.

We can then find a permutation matrix P such that $\rho_2^k(Pv) = (1a, 1b, 0c, 0d)^T$ or $(1a, 1b, 1c, 1d)^T$. In the latter case, multiplying by a suitable diagonal matrix (i.e., multiplying individual entries by appropriate powers of i) makes all second digits zero, yielding $(10,10,10,10)^T$.

In the former case, multiplying by a diagonal matrix makes the second digits of the odd entries zero, yielding $(10,10,0c,0d)^T$. Applying CK (equivalently, applying K to the pair of odd entries) produces $(00,01,0c,0d)^T$ or $(01,00,0c,0d)^T$. Since multiplication by CK preserves unit vectors, this is the residue of a unit vector with lde $k-1$. When $k > 1$, its $(1,k{-}1)$ residue must have the form $(1,1,0,0)^T$ up to a permutation, so $c = 1$ and $d = 0$.

When $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ for $l, m \in \{0,1,2,3\}$, up to a permutation. □

Lemma IV.7. *Let u, v be unit vectors in $\mathbb{D}[i]^4$ with lde $k > 0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized permutation matrix Q such that $\rho_1^k(Q[u|v])$ equals one of*

$$\begin{bmatrix}1&1\\1&1\\0&0\\0&0\end{bmatrix},\ \begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\1&0\\1&0\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\0&1\\0&1\end{bmatrix},\ \textit{or}\ \begin{bmatrix}1&1\\1&1\\1&1\\1&1\end{bmatrix}.$$

Proof. By Lemma IV.6, each column can have only an even number of 1s. The only missing case is

$$\begin{bmatrix}1&0\\1&1\\0&1\\0&0\end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(u_j^\dagger v_j) = 1$, contradicting $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices at which both u and v have $(1,k)$-residue equal to 1 must be even. □

Proof of Lemma IV.5. The rows and columns of a unitary matrix are unit vectors, and all row (column) vectors are pairwise orthogonal. Lemmas IV.6 and IV.7 force the pattern to be (ii)–(vi) when $k > 0$. When $k = 0$, the only unitaries with lde 0 are the generalized permutations, which have pattern (i). When $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ up to a permutation, so the residue cannot have four 1s in a column. Hence patterns (iv), (v), and (vi) occur only when $\mathrm{lde}(A) > 1$. □

Lemma IV.8. *For any $A \in \mathcal{CCS}$ with lde $k > 0$, there exist generalized permutations L, R, L', R' such that one of the following holds:*

- *$lde(K_1LAR) \le k-1$;*
- *$lde(LARK_1) \le k-1$;*
- *$lde(L'K_1LARR'K_1) \le k-1$;*
- *$lde(K_1L'K_1LARR') \le k-1$;*
- *$lde(L'CKLAR) \le k-1$ (only when $lde(A) = 1$).*

Each inequality is in fact an equality, but we do not need this. The proof proceeds by case analysis on $\rho_1^k(A)$. Since $k > 0$, Case (i) is excluded, so we begin with Case (ii).

1. Case ii

If $k > 1$, by Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 0d & 0e \\ 00 & 0\bar{c} & 0f & 0g \end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and b, c, d, e, f, g are indeterminates (henceforth we use indeterminates in residue matrices without explicit declaration). We claim $a = 0$ and $b = c = 1$. Taking the inner product of the first two columns gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

The inner product of the second and third rows gives $c = 1$; similarly, $b = 1$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix}.$$

Moreover, since $k > 1$ and the third and fourth rows (columns) have lde $k-1 > 0$, Lemma IV.6 gives $d = f$, $d = e$, and $e = g$, so $d = e = f = g$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0d \\ 00 & 00 & 0d & 0d \end{bmatrix}.$$

Applying K_1S_1 on the left reduces this to Case (iv).

If $k = 1$, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \end{bmatrix},$$

As before, $a = 0$. Applying $X_0 \cdot CK \cdot X_0$ on the left decreases the lde.

Remark IV.9. The two generalized permutations used to refine the pattern can be chosen to be CS-free. This is because the $(2,k)$-residues of the two rows (columns) with even entries are stable under multiplication by i, so we can insert appropriate powers of i at suitable positions to ensure the generalized permutation is CS-free.

2. Case iii

If $k > 1$, by Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix},$$

The inner product of the first two columns gives $a = 0$. The inner product of the second and third rows gives $c = 0$; similarly, $b = 0$. Now $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying K_1S_1 on the left reduces this to Case (vi).

If $k = 1$, by Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, $a = 0$ and $d = 0$. Applying K_1 on the left decreases the lde.

3. Case iv

This case occurs only when $k > 1$. By Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

The inner product of the first two columns gives $a = 0$; similarly, $b = c = 0$. Since the third row has lde $k-1 > 0$, either $d = e = f = 1$ or exactly one of d, e, f is 1 and the rest are 0. In the latter case, up to a column permutation, we may assume $d = 1$ and $e = f = 0$. In both cases, right multiplication by K_1 decreases the lde.

Remark IV.10. As in Remark IV.9, the left generalized permutation can be chosen to be CS-free.

4. Case v

This case occurs only when $k > 1$. By Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix},$$

We claim $a = 0$ and $b = c = e = 1$. The inner product of the first two rows gives $a = 0$. The inner product of the first and third rows gives $b = 1$; similarly, $c = 1$. The inner product of the second and third columns gives $e = 1$; similarly, $f = 1$. Now the pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix},$$

Applying K_1 on the left reduces this to Case (iv).

5. Case vi

This case occurs only when $k > 1$. By Lemma IV.6, up to left and right multiplication by generalized permutations, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1j \end{bmatrix},$$

The inner product of the first two columns shows that a, b, c cannot all be 1. Up to a row permutation, we may assume $a = 0$, which gives $b = c$; similarly, $d = g$.

If $b = c = 1$ and $d = g = 1$, then $\mathrm{pat}(A)$ is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1j \end{bmatrix},$$

The inner product of the first and third columns gives $f = \bar{e}$; similarly, $h = \bar{e}$ and $j = \bar{f}$. Now $\mathrm{pat}(A)$ is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We show this pattern is impossible by examining the $(3, k)$-residue. Up to multiplication by powers of -1 on appropriate rows and columns, the $(3, k)$-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11p \\ 101 & 11y & 1ev & 1\bar{e}q \\ 101 & 11z & 1\bar{e}w & 1er \end{bmatrix}.$$

The inner products of the first column with the second and third columns give

$$101 + 10x + 11y + 11z = 000 \implies x \oplus y \oplus z = 1,$$
$$101 + 11u + 1ev + 1\bar{e}w = 000 \implies u \oplus v \oplus w = 1.$$

The inner product of the second and third columns gives

$$101 + 11(u \oplus \bar{x}) + (1ev + 01e + 00\bar{y}) + (1\bar{e}w + 01\bar{e} + 00\bar{z}) = 000 \implies x \oplus y \oplus z \oplus u \oplus v \oplus w = 1, \text{ a contradiction.}$$

If $b = c = 0$, the inner product of the first and third rows gives $e = h$; similarly, $f = j$. So $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

The inner product of the first and third columns gives $d \oplus e \oplus f = 0$, so at least one of d, e, f is 0. Up to a row permutation, we may assume $d = 0$. Then, up to permutations of rows and columns, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

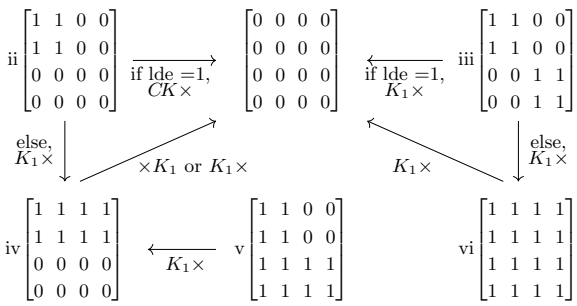

FIG. 1. Transitions between cases. The annotation $K_1\times$ ($\times K_1$) on an arrow means that if A has the pattern at the tail, then K_1LAR ($LARK_1$) has the pattern at the head for suitably chosen generalized permutations L and R. Similarly for other annotations. Note that CK decomposes into a circuit with two K gates as in Equation (1).

The inner product of the first and third columns implies $e = f$, so left multiplication by K_1 reduces the lde.

The case $d = g = 0$ is similar.

This exhausts all cases. The transitions are summarized in Figure 1.

Proof of Theorem IV.1. Let $k = \mathrm{lde}(A)$. By Lemma IV.8, and merging adjacent generalized permutations as needed, there exist generalized permutations $L_1, L_2, \ldots, L_p, R_1, R_2, \ldots, R_q$ such that

$$\mathrm{lde}(L_pK_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_q) = 0, \text{ i.e.,}$$

there exists a generalized permutation P such that

$$L_pK_1 \cdots L_2K_1L_1AR_1K_1R_2K_1 \cdots R_q = P.$$

The rkc of the left-hand side is $\leq 2k$. Therefore

$$A = L_1^{-1}iK_1L_2^{-1}iK_1 \cdots L_p^{-1}PR_q^{-1}iK_1 \cdots R_2^{-1}iK_1R_1^{-1}.$$

The rkc of the right-hand side is at most $2k$. After merging adjacent generalized permutations, the number of generalized permutations is at most $2k + 1$. Each generalized permutation is implementable by at most 9 gates and at most 1 CS gate, up to a global phase. This yields a circuit implementing A with at most $2k$ K gates, at most $2k + 1$ CS gates, and at most $9(2k + 1) + 2k$ gates up to a global phase. □

V. EXPERIMENT

We implemented our algorithm in Haskell and compared it with the CS-optimal synthesis algorithm of Glaudell et al. [17]. Unitaries with lde values in a given range were randomly generated and fed to both algorithms; the output circuits were compared on rcs and rkc. The output of their algorithm has the form of Equation (2) with optimal CS-count. We augmented their algorithm with one additional step to minimize the K-count: for each C_i, we replace it with an equivalent circuit of minimum K-count (which may be 0, 1, or 2). We randomly generated 1,346 two-qubit Clifford+CS operators with lde ranging from 2 to 58; data are available in Section VI. Figure 2 shows the distribution of lde values, and Figure 3 shows the distribution of the relative K-count reduction.

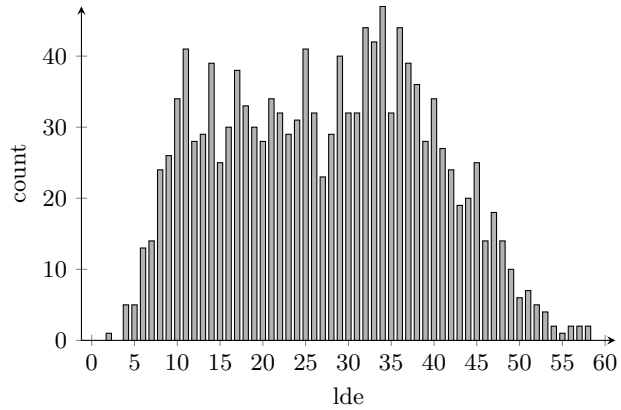

FIG. 2. Distribution of the least denominator exponent (lde) across the 1,346 test operators. The lde ranges from 2 to 58, with a mean of 27 and a median of 28.

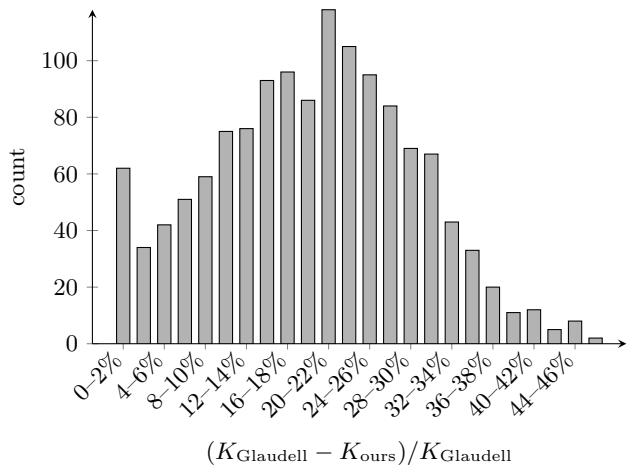

FIG. 3. Distribution of the relative K-count reduction $(K_{\mathrm{Glaudell}} - K_{\mathrm{ours}})/K_{\mathrm{Glaudell}}$, binned in 2 % increments. Our method improves on Glaudell et al.'s K-count in 95.5 % of cases, with a mean reduction of 19.4 %, a median of 19.8 %, and a maximum of 47.7 %.

For every operator, our CS-count and that of Glaudell et al. are identical, so our algorithm is also CS-optimal in practice. The K-count is strictly reduced in 95.5% of cases; see Figure 3.

The inequalities $\mathrm{lde} - 2 \leq CS\text{-count} \leq K_{\mathrm{ours}}$ hold for every operator. In particular,

- $\mathrm{lde} \leq CS$-count: 1,324 cases (98.4%),
- $\mathrm{lde} > CS$-count: 22 cases (1.6%).

In the 22 cases, the deficit is usually 1, not 2.

A. Source of improvement

Two tight linear relationships emerge from the data. First, the K-count of Glaudell et al. is well approximated by twice the CS-count:

$$K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}.$$

The difference lies in $\{-4, -3, -2, -1, 0, 1, 2\}$, with 93.5% of values in $\{-4, \ldots, 0\}$. Second, our K-count is well approximated by twice the lde:

$$K_{\text{ours}} \approx 2 \cdot \text{lde}.$$

The difference lies in $\{-2, -1, 0\}$, with $\{-1, 0\}$ accounting for 80.5% of cases. These relationships are also reflected in the Pearson correlations: lde vs. K_{ours} has correlation 0.9996, while CS-count vs. K_{Glaudell} has correlation 0.9991. Since $K_{\text{Glaudell}} \approx 2 \cdot CS$-count and $K_{\text{ours}} \approx 2 \cdot \text{lde}$, the absolute K-count saving is approximately $2 \cdot (CS\text{-count} - \text{lde})$. The relative reduction is roughly stable across lde values, hovering around 19–21% for lde between 5 and 40, and tapering off at the extremes (possibly due to small sample sizes).

VI. CODE AND DATA AVAILABILITY

Code and data are available from [7].

VII. CONCLUSION AND FUTURE WORK

We have given an efficient, near-optimal algorithm for the exact synthesis of two-qubit Clifford+CS circuits from a given unitary matrix. The theoretical worst-case CS-count is 4 times the optimal, but in practice our method always achieves the optimal CS-count. The theoretical worst-case K-count is 2 times the optimal, and in practice our method consistently achieves a better K-count than the best previously known algorithm.

The algorithm of Li et al. [30] produces circuits whose T-count is at worst 10 times the optimal. We believe our method can be adapted to improve this bound.

VIII. ACKNOWLEDGEMENTS

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- creating four references [12, 15, 28, 29] and their one-line summary in the related work section.
- improving the wording of the paper.
- generating Haskell code: the ParseData.hs (for parsing data and drawing distribution diagrams) and CliffordCS.hs (the implementation of Glaudell et al.'s algorithm [17]).
- drafting Section VI, with proper guide from the authors (later revisions are made).

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